Adriana Lucía Ruiz Rizzo

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Academic Career

Education

2014.04 – 2018.06	Ph.D. Systemic Neuroscience, GSN, LMU Munich, Germany Dissertation: Visual Processing Speed in the Aging Brain Date of Defense: 27.06.2018 Supervisors: Prof. Dr. Kathrin Finke, PD Dr. Christian Sorg, Prof. Dr. Hermann J. Müller
2011.09 – 2013.08	M.Sc. Cognitive and Clinical Neuroscience, Maastricht University, Netherlands
2021.01 – 2021.12	Postgraduate Degree in Applied Statistics, University Los Libertadores, Colombia
2003.07 – 2008.05	B.A. in Psychology, University of Antioquia, Colombia

Experience

2022.04 – Present	Postdoctoral researcher, Jena University Hospital, leading the project 'Brain connectivity underpinnings of visual attention dysfunctions in mild cognitive impairment.' *[Maternity leave: 2023.04 – 2024.06]*
2021.04 – 2022.03	Postdoctoral Research Fellow, Department of Psychology, LMU Munich, Project 'Visual processing speed and functional and structural connectivity in older adults with subjective cognitive decline.'
2019.10 – 2021.03	Postdoctoral Research Fellow, Department of Psychology, Institute of Gerontology, Wayne State University, Detroit, MI, U.S.A
2018.07 – 2019.09	Postdoctoral Research Fellow, Department of Psychology, LMU Munich, Germany
2012.10 – 2013.08	Master's Research Intern, Donders Institute, Radboud University Nijmegen, the Netherlands (Master's thesis)
2008.04 – 2011.08	Graduate Research Assistant, Grupo de Neurociencias de Antioquia (GNA) research group, Universidad de Antioquia, Colombia
2008.04 – 2011.08	Clinical Neuropsychologist, GNA
2007.05 – 2007.11	Undergraduate Research Intern, Laboratorio de Neurociencias, Universidad del Zulia, Venezuela
2006.08 – 2007.05	Undergraduate Research Assistant, GNA research group, Universidad de Antioquia, Colombia

Teaching

2018.02 – 2019.09	Lecturer and Tutor, M.Sc. program in Neuro-Cognitive Psychology and M.Sc. program in Psychology, LMU Munich, Germany. Subjects: Neuropsychological methods; Cognitive neuroscience of aging; Specific Methods in Neurocognitive Psychology: Functional Magnetic Resonance Imaging
2010.09 – 2011.09	Lecturer, Department of Psychology, Universidad de Antioquia, Colombia. Subjects: Theories of the mind; Biopsychology;

	Neuropsychology (80 hours per subject per academic term)
2008.10 – 2010.06	Tutor, Faculty of Medicine, Universidad de Antioquia, Colombia. Subjects: Memory and Amnesia; Language and Aphasia (30 hours in total per academic term)

Supervision and Mentoring

2023.03 – Present	Supervisor doctoral candidate Universitätsklinikum Jena, Germany Ph.D. candidate supervised: Hannah Klink, M.Sc.
2018.07 – 2023.07	Research Internships Supervisor, M.Sc. program in Neuro-cognitive Psychology (NCP), Department of Psychology, LMU Munich, Germany Students supervised: - Sebastian Schneider, Ph.D. *Project: A Frequency-Band Dependent Analysis of Age-Related Decreased Cingulo-Opercular Network Functional Connectivity and its Association with Visual Processing* - Svenja Küchenhoff, Ph.D. *Project: Visual processing speed links with frequency-band-specific inter-network functional connectivity between the cingulo-opercular network, the right frontoparietal network, and visual networks* - Oliver Kohl, Ph.D. *Project: Effects of Age and Visual Attention on Metastability of functional connectivity* - Mary-Ann Kubre, M.Sc. *Project: Functional Anatomical Research in Transgender Persons* - Daniela Marrero-Polegre, M.Sc. *Project: Visual processing speed and subjective cognitive complaints in healthy aging* - International visiting undergraduate student supervised: Neha Sharma, B.Sc., M.D. Case Western Reserve University, U.S.A. *Project: The relationship(s) between Sleep Quality, brain structure, and the parameters of visual attention*
2018.07 – 2023.07	NCP master's theses supervised: - Age Effects on the Basal Forebrain Cholinergic System and Links to Visual Processing Speed in a Non-Demented Sample: A Voxel-Based Morphometry Study. Student: Dorontina Ismajli (2019) - Visual Processing Speed and Intrinsic Functional Connectivity of Cholinergic Nucleus Basalis Meynert and Noradrenergic Locus Coeruleus. Student: Sebastian Schneider (2019) - Association between Fatigue- and Sleep-Relevant Brain Networks and Visual Attention Parameters in Multiple Sclerosis. Student: Daniela Marrero Polegre (2023)

Third-party funding

2023.03 – 2026.03	IZKF Advanced Medical Scientist-Programm 2022, Universitätsklinikum Jena, Germany (150,000 €)
2019.10 – 2022.03	LMU Outgoing Research Fellowship for outstanding postdoctoral researchers, LMU Munich, Germany (159,000 €)

2008.11 – 2009.05	Implementation project Training Course on Executive Functions, Mayoralty of Medellin, Colombia (12,000 €)

Stipends

2017.08 – 2017.12	Stipend from the Department of Psychology, LMU Munich (8,000 €)
2017.05 – 2017.07	Stipend from the Graduate School of Systemic Neurosciences, LMU Munich (4,800 €)
2014.04 – 2017.04	PhD funded under EU Marie Curie Actions FP7 (International Training Network)
2016.06 – 2016.06	FENS-IBRO/PERC Travel Grant, Denmark (750 €)
2011.09 – 2013.08	Colfuturo Scholarship for Master's Studies, Colombia (49,000 €)
2011.09 – 2013.08	Enlazamundos Scholarship for Master's Studies, Colombia (5,000 €)
2007.05 – 2007.11	Studentship, EC-ALFA Programme, Colombia (6,000 €)

Awards

2020.09	SCHUHFRIED Science Award for the work on "Alertness training increases visual processing speed in healthy older adults". Austria
2020.03	1st Place Poster Presentation Prize (postdoctoral level) at the 10th Annual Graduate and Postdoctoral Research Symposium for the poster "Is recognizing a tree more difficult than recognizing a forest? Scenes' object load modulation of memory recognition in healthy older adults", Wayne State University. United States
2013.10	Alejandro Ángel Escobar National Award in Biological Sciences and Physics for Research of Genetic Alzheimer's Due to E280A Mutation in the presenilin-1 (PS1) Gene in Families of Antioquia. Colombia
2008.11	Outstanding Undergraduate Researcher Award. Mayoralty of Medellin. Colombia
2008.08	Talent Young Woman in Science and Technology Award. Mayoralty of Medellin. Colombia
2005.10	Best Advanced Student in Psychology 2004, University of Antioquia. Colombia

Academic communication

2022	Structural MRI connectivity and diet in the DELCODE study. 3rd International conference on cognitive reserve in dementia and other disorders (ResDem 2022), LMU Munich. Munich, Germany. November 19, 2022 (Program)
2022	Visual Processing Speed in the Aging Brain. Research Colloquium of the Institute of Gerontology, Wayne State University (U.S.A.). September 27, 2022 (Recording)
2020-2021	Functional connectivity between right frontoparietal and visual networks is linked to visual processing speed. Neuromatch Conference 3.0, October 29, 2020 (Recording) and DataBlitz at the Cognitive Neuroscience Society (CNS) meeting, February 2021 (Recording)

Scientific publications

2026 [Preprint] **Ruiz-Rizzo AL**, Schrenk SJ, Brodoehl S, Frahm C, Gaser C, Herbsleb M, ... & Finke K. Physical exercise increases the NODDI-derived neurite density index across white matter tracts in healthy older adults: results from the FIT4BRAIN randomized controlled trial. medRxiv, 2026-08. https://doi.org/10.64898/2026.08.31.26361810

2026 Schrenk SJ, Bang, C, Best L, Dost T, Flor S, Frahm, C, …, **Ruiz-Rizzo AL**, ..., & Finke K. Impact of an online-guided physical activity intervention on cognition, resting-state brain connectivity, and the gut microbiome in healthy older adults—a randomized controlled trial. GeroScience, 1-23. https://doi.org/10.1007/s11357-026-02324-6

2025 Balßuweit K, Bublak P, Finke K, **Ruiz-Rizzo AL**, Wagner F, Klingner C, & Brodoehl S. A multimodal MRI framework employing machine learning for detecting beginning cognitive impairment in Parkinson's disease. Frontiers in neuroscience, 19, 1689302. https://doi.org/10.3389/fnins.2025.1689302

2025 Schirge PM, Perneczky R, Taoka T, **Ruiz-Rizzo AL**, Ersoezlue E, Forbrig R, ... & Rauchmann BS. Perivascular space and white matter hyperintensities in Alzheimer's disease: associations with disease progression and cognitive function. Alzheimer's research & therapy, 17(1), 1-20. https://doi.org/10.1186/s13195-025-01707-9

2025 Gerner K, Bublak P, Finke K, Schrenk S, **Ruiz-Rizzo AL**, Wagner F, ... & Brodoehl S. Visual processing capacity and cognitive decline in Parkinson's disease. BMJ Neurology Open, 7(1), e000953. https://doi.org/10.1136/bmjno-2024-000953

2024 **Ruiz-Rizzo AL**, Finke K, Damoiseaux JS, …, Perneczky R, Rauchmann BS. Fornix fractional anisotropy mediates the association between Mediterranean diet adherence and memory four years later in older adults without dementia. Neurobiology of Aging, 136, 99-110. https://doi.org/10.1016/j.neurobiolaging.2024.01.012

2023 Marrero-Polegre D, Finke K, Roaschio N, Haupt M, Reyes-Moreno C, **Ruiz-Rizzo AL**. Lower visual processing speed relates to greater subjective cognitive complaints in community-dwelling healthy older adults. Frontiers in Psychiatry, 14, 1063151. https://doi.org/10.3389/fpsyt.2023.1063151

2023 Nuernberger M, Finke K, Nuernberger L, **Ruiz-Rizzo AL**, Gaser C, Klingner C, ... & Brodoehl S. Visual stimulation by extensive visual media consumption can be beneficial for motor learning. Scientific reports, 13(1), 22056. https://doi.org/10.1038/s41598-023-49415-4

2023 Schrenk SJ, Brodoehl S, Flor S, Frahm C, Gaser C, Hamdan RA, ... **Ruiz-Rizzo AL**, …, & Finke K. Impact of an online guided physical activity training on cognition and gut-brain axis interactions in older adults: protocol of a randomized controlled trial. Frontiers in Aging Neuroscience, 15, 1254194. https://doi.org/10.3389/fnagi.2023.1254194

2022 **Ruiz-Rizzo AL**, Viviano RP, Daugherty AM, Finke K, Müller HJ, Damoiseaux JS. Subjective cognitive decline predicts lower cingulo-opercular network functional connectivity in individuals with lower neurite density in the forceps minor. NeuroImage, 263, 119662. https://doi.org/10.1016/j.neuroimage.2022.119662

2022 **Ruiz-Rizzo AL**, Bublak P, Kluckow S, Finke K, Gaser C, Schwab M, Güllmar D, Müller HJ, Witte OW, Rupprecht S. Neural distinctiveness of fatigue and low sleep quality in multiple sclerosis. European Journal of Neurology, 29(10), 3017-3027. https://doi.org/10.1111/ene.15445

2022 **Ruiz-Rizzo AL**, Archila-Meléndez ME, González Veloza JJF. Predicting the probability of finding missing older adults based on machine learning. Journal of Computational Social Science, 1-19. https://doi.org/10.1007/s42001-022-00171-x

2022 **Ruiz-Rizzo AL**, Pruitt PJ, Finke K, Müller HJ, Damoiseaux JS. Lower-Resolution Retrieval of Scenes in Older Adults With Subjective Cognitive Decline. Archives of Clinical Neuropsychology, 37(2), 408–422. https://doi.org/10.1093/arclin/acab061

2021 Küchenhoff S, Sorg C, Schneider SC, Kohl O, Müller HJ, Napiórkowski N, Menegaux A, Finke K*, **Ruiz-Rizzo AL***. Visual processing speed is linked to functional connectivity between right frontoparietal and visual networks. European Journal of Neuroscience, 53, 3362-77. https://doi.org/10.1111/ejn.15206 * *equal contribution*.

2021 Penning MD*, **Ruiz-Rizzo AL***, Redel P, Müller HJ, Salminen T, Strobach T, Behrens S, Schubert T, Sorg C, Finke K. Alertness training increases visual processing speed in healthy older adults. Psychological Science, 32(3), 340-353. https://doi.org/10.1177/0956797620965520 * *equal contribution*.

2020 Haupt M, **Ruiz-Rizzo AL**, Sorg C, Finke K. Right-lateralized fronto-parietal network and phasic alertness in healthy aging. Scientific Reports, 10, 4823. https://doi.org/10.1038/s41598-020-61844-z

2020 Menegaux A, Bäuerlein FJB, Vania A, Napiorkowski N, Neitzel J, **Ruiz-Rizzo AL**, Müller HJ, Sorg C, Finke K. Linking the impact of aging on visual short-term memory capacity with changes in the structural connectivity of posterior thalamus to occipital cortices. Neuroimage, 208, 116440. https://doi.org/10.1016/j.neuroimage.2019.116440

2020 **Ruiz-Rizzo AL**, Beissner F, Finke K, Müller HJ, Zimmer C, Pasquini L, Sorg C. Human subsystems of medial temporal lobes extend locally to amygdala nuclei and globally to an allostatic-interoceptive system. Neuroimage, 207, 116404. https://doi.org/10.1016/j.neuroimage.2019.116404

2019 Haupt M, **Ruiz-Rizzo AL**, Sorg C, Finke K. Phasic alerting effects on visual processing speed are associated with intrinsic functional connectivity in the cingulo-opercular network. Neuroimage, 196, 216-26. https://doi.org/10.1016/j.neuroimage.2019.04.019

2019 Menegaux A, Napiorkowski N, Neitzel J, **Ruiz-Rizzo AL**, Petersen A, Müller HJ, Sorg C, Finke K. Theory of visual attention thalamic model for visual short-term memory capacity and top-down control: Evidence from a thalamo-cortical structural connectivity analysis. Neuroimage, 195, 67-77 https://doi.org/10.1016/j.neuroimage.2019.03.052

2019 **Ruiz-Rizzo AL**, Sorg C, Napiorkowski N, Neitzel J, Menegaux A, Müller HJ, Vangkilde S, Finke K. Decreased cingulo-opercular network functional connectivity mediates the impact of aging on visual processing speed. Neurobiology of Aging, 73, 50-60. https://doi.org/10.1016/j.neurobiolaging.2018.09.014

2018 **Ruiz-Rizzo AL**, Neitzel J, Müller HJ, Sorg C, Finke K. Distinctive correspondence between separable visual attention functions and intrinsic brain networks. Frontiers in Human Neuroscience, 12, 89 https://doi.org/10.3389/fnhum.2018.00089

2017 **Ruiz-Rizzo AL**, Bublak P, Redel P, Grimmer T, Müller HJ, Sorg C, Finke K. Simultaneous object perception deficits are related to reduced visual processing speed in amnestic mild cognitive impairment. Neurobiology of Aging, 55, 132-42. https://doi.org/10.1016/j.neurobiolaging.2017.03.029

2014 Londono AC, Castellanos FX, Arbelaez A, **Ruiz A**, Aguirre-Acevedo DC, Richardson AM, Easteal S, Lidbury BA, Lopera F. An 1H-MRS framework predicts the onset of Alzheimer's disease symptoms in PSEN1 mutation carriers. Alzheimer's & Dementia, 10(5), 552-61. https://doi.org/10.1016/j.jalz.2013.08.282

2013 Quiroz YT, Stern CE, Reiman EM, Brickhouse M, **Ruiz A**, Sperling RA, Lopera F, Dickerson BC. Cortical atrophy in presymptomatic Alzheimer's disease presenilin 1 mutation carriers. Journal of Neurology Neurosurgery & Psychiatry, 84(5), 556-61. https://dx.doi.org/10.1136/jnnp-2012-303299

2011 Quiroz YT, Ally BA, Celone K, McKeever J, **Ruiz-Rizzo AL**, Lopera F, Stern CE, Budson AE. Event related potential markers of brain changes in preclinical familial Alzheimer disease. Neurology, 2, 77(5), 469-75. https://doi.org/10.1212/WNL.0b013e318227b1b0

2010 Quiroz YT, Budson AE, Celone K, **Ruiz A**, Newmark R, Castrillón G, Lopera F, Stern CE. Hippocampal Hyperactivation in Presymptomatic Familial Alzheimer's Disease. Annals of Neurology. 68(6): 865-75. https://doi.org/10.1002/ana.22105

2009 **Ruiz-Rizzo AL**, Tirado V, Moreno C, Aguirre-Acevedo DC, Murillo O, Lopera F. [Comparison of neuropsychological performance according to the age of onset in subjects with Parkinson's disease and Parkinsonism]. Revista de Neurología, 49(3), 123-30 PMID: 19621306. https://doi.org/10.33588/rn.4903.2008475

2009 **Ruiz AL**. [Neuropsychological Alterations in patients with Major Depression]. Acta Neurológica Colombiana, 25(2), 75-83. https://actaneurologica.com/index.php/anc/article/view/1548

2008 **Ruiz Rizzo AL**, Lopera F. [Explicit memory and major depression]. Iatreia, 21(2), 177-85. https://doi.org/10.17533/udea.iatreia.4479

2024	[Book chapter] **Ruiz-Rizzo AL,** Finke K, & Archila-Meléndez ME (2024). Diffusion tensor imaging in alzheimer's studies. In Biomarkers for Alzheimer's Disease Drug Development (pp. 105-113). New York, NY: Springer US. https://doi.org/10.1007/978-1-0716-3774-6_8

Professional service

Peer reviewer	Brain, Alzheimer's research & therapy, Human Brain Mapping, Cerebral Cortex, Neuroimage, Neuroimage: Clinical, Neurobiology of Aging, Alzheimer's & Dementia, Biological psychiatry, Cognitive neuroscience and neuroimaging, Journal of Alzheimer's Disease, Frontiers in Aging Neuroscience, International journal of geriatric psychiatry, Canadian journal on aging, Brain and behavior, Acta psychologica, European journal of neuroscience, Journal of the International Neuropsychological Society, BMC geriatrics, Aging & mental health, Journal of cognitive neuroscience, Journal of magnetic resonance imaging, Psychological medicine, PloS one, Consciousness and cognition, Frontiers in psychology, Journal of integrative neuroscience, Gut microbiome.
Academic editor	Journal of Aging Research and BMC Geriatrics